

How to spot a...

Class 701

**The even more
comprehensive guide**

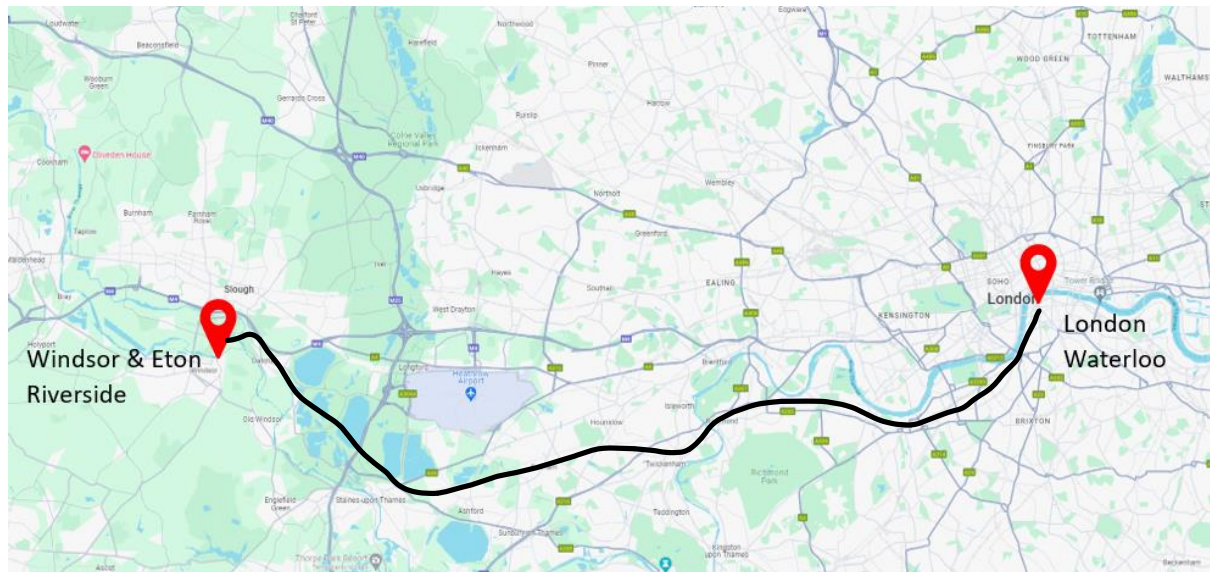
Version 2 - July 2024 Edition

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Locations

Class 701s currently operate ONLY* on the South Western Main Line (SWML) out of London Waterloo to Winsor & Eton Riverside, near Slough in Berkshire.



*To avoid ambiguity, they ONLY operate on this line. They **DO NOT** operate on any other lines such as the Cambridge - Kings Cross line.

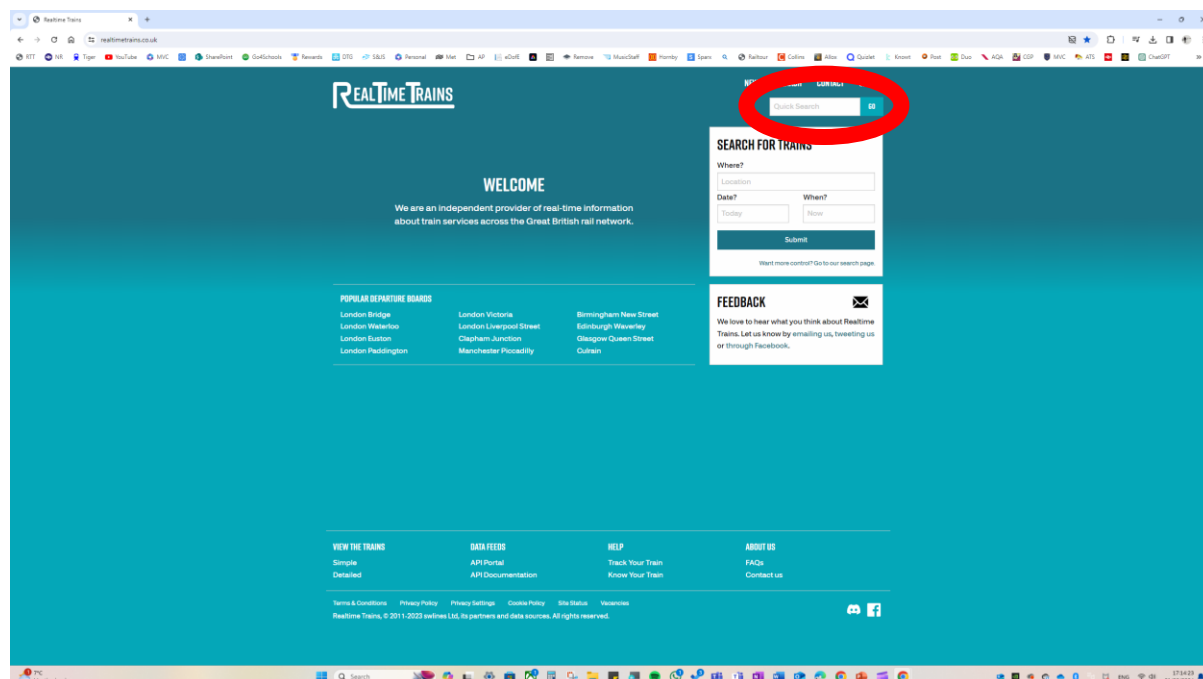
Diagrams

The class 701's have recently been introduced, so only a few sets of 701s are operating on the SWML. You can find out if a 701 is operating a diagram by using the RealTimeTrains website, using the train's headcode.

Website: <https://www.realtimetrains.co.uk/>

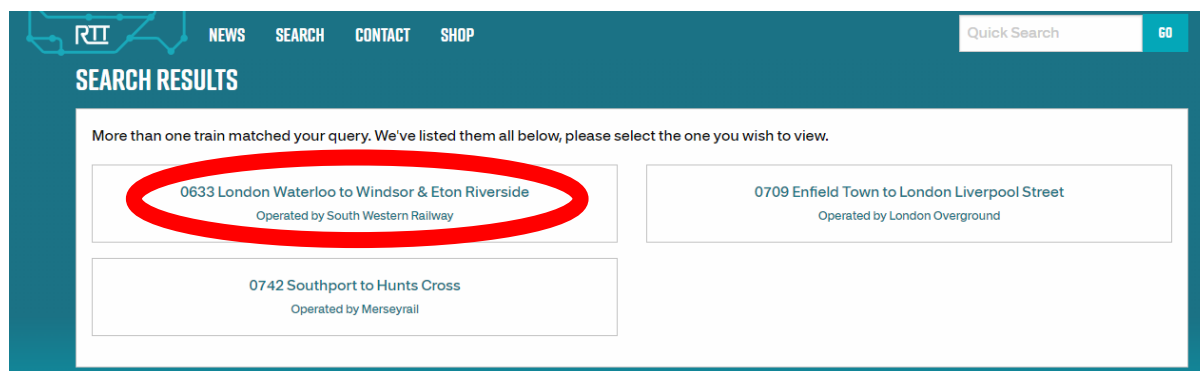
Step 1: Go to the website.

Step 2: In the 'quick search' tab, enter the train headcodes as shown below.



Step 3: Click on 'GO'

Step 4: Some trains in the UK have the same headcode, so it will bring up a box to choose the correct service. In this case there are multiple trains with headcode 2U09, so we need to click on: '0633 London Waterloo to Windsor & Eton Riverside'



Step 5: And voila! You've found your service which is a 701!!...

[NEWS](#)
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0633 LONDON WATERLOO TO WINDSOR & ETON RIVERSIDE
DEPARTING TODAY

OPERATED BY
SOUTH WESTERN RAILWAY

10 coaches

Location	Pl	Planned		Realtime		
		Arr	Dep	Arr	Dep	Dly
WAT London Waterloo	23		0633		0636	+3
VXH Vauxhall	3	0637	0637	0639	0640	+3
CLJ Clapham Junction	6	0642	0643	0644	0645	+2
PUT Putney	4	0646	0647	0648	0649	+2
RMD Richmond	1	0652	0653	0654	0655	+2
TWI Twickenham	5	0656	0657	0657	0658	+1
WTN Whitton	2	0700	0700	0700	0701	+1
FEL Feltham	2	0704	0704	0704	0705	+1
AFS Ashford (Surrey)	2	0708	0708	0708	0709	+1
SNS Staines	2	0711	0712	0711	0712	
WRY Wrybury	2	0716	0717	0715	0716	-1
SNY Sunnymeads	2	0719	0720	0718	0720	
DAT Datchet	2	0723	0724	0722	0723	-1
WNR Windsor & Eton Riverside	2	0729		0727		-2

Go to detailed version

Standard class only seating

Oak Frames Direct

Oak Frames Direct
Open >

Step 6: At the bottom of the page, click 'Go to detailed version'

Now you can see a full outline of the service.

You can also see lots of detailed data, which is explained on the next page...

How to read RealTimeTrains - The Professional Way

Service departure date, you can view services up to 7 days in the past and 180 days in the future

Service diagram, compatible services only, see [here](#) for details

Service identification information

UID - Unique Identification, needed as some trains have same 'headcodes' in the uk

Identity - the 'headcode'

TSC - Train Service Codes

Headcode - not to be confused with the other 'headcode', see page 7.

Dates when the service operates from
EWD = Every Week Day
SX = Operates Sun-Fri
SuX - Operates Mon-Sat
SSuX - Mon-Fri
M = Mon
Tu = Tuesday
W = Wed
Th = Thurs
F = Fri
S = Sat
Su = Sunday
O = Only e.g. FO = Fri Only

What class of train is operating the service, compatible services only, see [here](#) for details

Service information, e.g. seating, catering, weather it operates without a guard etc.

TRUST - Train Running Under System TOPS. Part of Network rail's open data feed which the Real Time Trains website runs off of. It monitors the progress of trains and tracks delays.

Date and time the service was activated under TRUST.

'Headcode'

Departure time

Service origin

Service destination

Service operator

Service facilities:

TRAIN FACILITIES

- Bicycle spaces
- Accessible space
- Toilets
- Air Conditioning

COACH FACILITIES

- Bicycle spaces
- Accessible space
- Buffet car
- Quiet coach
- Sleeper car
- First class
- Luggage storage

ON-BOARD POWER AND CONNECTIVITY

- WiFi (wireless internet access)
- Power sockets, not guaranteed at every seat
- USB sockets (USB type A)
- Entertainment (over WiFi)

See [here](#) for more details

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Date and time the service was activated under TRUST.

2U09 0633 LONDON WATERLOO TO WINDSOR & ETON RIVERSIDE

DEPARTING TODAY

OPERATED BY SOUTH WESTERN RAILWAY

10 coaches

Coach 1 Coach 2 Coach 3 Coach 4 Coach 5 Coach 6 Coach 7 Coach 8 Coach 9

701037

Train class category for the headcode, see page 7

It runs in Great Britain. TPS - Train Planning System

Shows what type of train it is (electric or diesel) and how fast it can go

Miles and chains	Location	Platform	GBTT		WTT		Realtime		Dly	Path	Line
			Arr	Dep	Arr	Dep	Arr	Dep			
0.00	London Waterloo [WAT]	23		0633		0633	0636%	+3	WR1		
0.40	Waterloo West Crossings					pass	0634½	pass	0637½	+3	WR1 DWF
1.29	Vauxhall [VXH]	3	0637	0637	0636½	0637½	0639%	0640%	+3		
1.78	Nine Elms Jn					pass	0639	pass	0641½	+2	DWF DW
2.50	Queenstown Road Battersea [QRB]	3				pass	0640	pass	0642½	+2	DW DWS
3.17	West London Jn					pass	0640½	pass	0643	+2	
3.74	Clapham Junction [CLJ]	6	0642	0643	0642	0643	0644%	0645%	+2	DWS DWS	
4.60	Wandsworth Town [WNT]	1				pass	0644½	pass	0647	+2	
5.72	Putney [PUT]	4	0646	0647	0646	0647	0648%	0649	+2		
7.07	Barnes [BNS]	4				pass	0649	pass	0651	+2	DWS DR
8.21	Mortlake [MTL]	2				pass	0650	pass	0652	+2	
9.03	North Sheen [NSH]	2				pass	0651	pass	0652½	+2	
9.57	Richmond [RMD]	1	0652	0653	0652	0653	0654	0654%	+1	DR DT	
10.66	St Margarets (London) [SMG]	3				pass	0655	pass	0656½	+1	
11.22	Twickenham [TWI]	5	0656	0657	0656	0657	0657½	0658%	+1		
11.49	Twickenham Jn					pass	0658	pass	0658½	+1	
12.43	Whitton [WTN]	2	0700	0700	0659½	0700	0700%	0701	+1		
13.03	Whitton Jn					pass	0701	pass	0702	+1	
13.35	Feltham Jn					pass	0701½	pass	0702½		
14.68	Feltham [FEL]	2	0704	0704	0703½	0704½	0704%	0705			
17.40	Ashford (Surrey) [AFS]	2	0708	0708	0708	0708½	0708%	0709			
19.02	Staines [SNS]	2	0711	0712	0711½	0712½	0711%	0712½		DT DW	
21.40	Wraybury [WRY]	2	0716	0717	0716	0717	0715	0716	-1		
22.48	Sunnymeads [SNY]	2	0719	0720	0719	0720	0718	0720			
23.63	Datchet [DAT]	2	0723	0724	0722½	0724	0722	0723	-1		
	1 min pathing allowance										
25.48	Windsor & Eton Riverside [WNR]	2	0729		0729		0727		-2		

Miles and chains

Service stops locations, if its white, it calls there, if the shady blue colour, it doesn't call there

Platform, if it goes red it means there has been a platform alteration.

Arrival and departure times.

Left is arrival time, red is departure time

GBTT is the public times, to the nearest minute

WTT (Working Time Table) are more accurate timings, to the nearest 30 seconds

Realtime timings are the service's actual timings, to the nearest 15 seconds.

Service delays

+ = minutes late

- = minutes early

Service routes, these have many abbreviations, so I can't put them all.

Lines towards London are called up lines, lines away from London are called down lines

Explanation of headcodes

Modern day headcodes consist of a single digit number, followed by a letter then a 2 digit number.

Old headcodes, sometimes called 'train reporting numbers' are still used, and they date back to old British Rail days. These show what route the train is travelling on. From [this](https://sremg.org.uk/headcodes/ehheadcodes/ehheadcodes01.html) website (<https://sremg.org.uk/headcodes/ehheadcodes/ehheadcodes01.html>), you can see that 58 is correctly 'Waterloo - Windsor via Richmond'. They were called the headcode as they were displayed on the 'head' of the train as seen below:



Modern day headcodes, such as 1A01, are assigned to every service. You can tell a lot about a train just from its headcode:

1. The first digit is the train class category.

0 represents a light engine movement (just a single locomotive travelling). An example could be 0V60, because the first digit is a zero.

1 represents an express train / fast train. 1 is also used for rail charters, usually when the 1 is followed by a Z.

2 represents a semi fast or slow passenger train (ordinary passenger), which frequently stops at stations.

3 is a priority ECS (empty coaching stock) train, parcels, or a seasonal / weather related vehicle used to spray the rails or weeds.

4 is a fast freight train.

5 is an empty coaching stock train not in passenger use.

6 is an aggregates train, building materials, or other freight which is slower than a class 4 train.

7 is even slower than a class train, usually doing no more than 45mph.

8 is often weather related vehicles, and very slow trains.

9 is high speed services such as channel tunnel passenger services, Class 9 headcode is also used for passenger services that are running with restrictions or out of gauge in places, that require special routing. The 9 headcode tells the signaller that this is a train with special requirements.

2. The second digit is the route identifier

Regionally, most routes will have their own letter. For example, the Kings Cross to Cambridge trains use a C. Different local routes will have a different letter to tell apart train services. Q is usually used for test trains, and Z is usually used for Short Term Plan workings which don't usually run. Z can be used for test trains, and charter trains.

3 and 4.

Digits 3 and 4 are incremental for standard passenger services, and quite simply, identify particular train services. For example 0900 Brighton to Victoria will have maybe 1A01, and then the 0911 may have 1A02. The last two digits just identify the service on that day.

The Working Time Table (WTT)

The working timetable (WTT) is the rail industry's version of the public national timetable

You can download them for the whole country here - <https://www.networkrail.co.uk/running-the-railway/the-timetable/working-timetable/>

They used to just be available as .pdf files but now they upload the .xlsx files which is very nice of them.

As an example, this is what it looks like...

[illegible]

This one's from Kings Cross to Welwyn Garden City.

I'm not going to detail on this, it's mostly self-explanatory but can get confusing as it has a lot of detailed information.

When you download them, you want to click on 0.1 WTT June 2024 - December 2024 which is the latest. You can download the mon-fri, the Saturday or the Sunday versions.

[This](#) is the book index to find which one you want which corresponds to the book file to download

Some helpful ones:

Kings Cross and Moorgate to Hertford North and Welwyn Garden City

Welwyn Garden City and Hertford to Royston and Grantham

Royston and Stansted Mountfitchet to Cambridge and Coldham Lane Jn

Waterloo to Feltham

Feltham to Windsor, Weybridge and Reading, Guildford to Wokingham

How to spot

Spotting a train is fairly self-explanatory. Just turn up at the station, at the time shown on real time trains, and see the train. You can use open train times to track the headcode of your train.

https://www.opentraintimes.com/maps/signalling/wat#T_WATRLMN

If you have any questions, you can get in touch by Whatsapp.

List of all services operated by class 701

NOTE: these diagrams can change frequently and at short notice, and may become out of date.

Services from Summer 2024 WTT, accurate as of 18 July 2024

They are very rare to spot, there is still only 1 train running at the moment.

Diagram 1:

5U09 0540 Wimbledon Park Depot Sdgs to London Waterloo (empty stock move)

2U09 0633 London Waterloo to Windsor & Eton Riverside

2U16 0753 Windsor & Eton Riverside to London Waterloo

2U19 0903 London Waterloo to Windsor & Eton Riverside

2U26 1023 Windsor & Eton Riverside to London Waterloo

5Y89 1128 London Waterloo to Wimbledon Park Depot Sdgs (empty stock move)

Useful Links

The Wikipedia page on the 701 has lots of useful information -

https://en.wikipedia.org/wiki/British_Rail_Class_701

Realtime trains for seeing detailed train times - <https://www.realtimetrains.co.uk/>

The open train times website's live rail maps are really useful, as well as really cool -

<https://www.opentraintimes.com/maps>

Glossary

SWML - South Western Main line, see page 3

Set or unit - another word for a single train

Timetable - the times that the train runs to, and what stations it stops at

Diagram - the train number (e.g 701 010) that is allocated to a timetable

Headcode - the code that each train has which identifies it (e.g 1A01)

WTT - Working Time Table - the railway time table, with more in depth data such as empty stock moves.

Empty stock moves - trains moving out of service, without any passengers.

Helpful abbreviations can be found here: <https://www.rail-record.co.uk/uk-railway-terms-glossary/>

Acknowledgements

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Train reporting number image https://www.wikiwand.com/en/Train_reporting_number

Credits

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Permission to reproduce all copyright material has not been applied for because I do not have the time. Sorry. If anyone has a massive problem I will be happy to rectify any omissions or acknowledgements.

This document has now been typeset in 'Aptos'. (p.s. much better than calibri)

